

$$-g\theta = l \frac{d^2\theta}{dt^2}$$

$$-g\theta = F = l\alpha$$

$$F = -kx = ma$$

- (e) Die vergelyking wat die EHB van 'n veer beskryf is $m \frac{d^2x}{dt^2} = -kx$. Toon aan dat die periode van die pendulum gegee word deur:

$$l \frac{d^2\theta}{dt^2} = -g\theta = l\alpha$$

The equation describing the SHM of a spring is given by $m \frac{d^2x}{dt^2} = -kx$. Show that the period of the pendulum is given by:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\alpha = \frac{a}{l}$$

$$a = -\omega^2 \theta$$

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$$\omega = \sqrt{\frac{g}{l}}$$

$$\omega = \frac{2\pi}{T}$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

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(2)

- (f) Beskou 'n dun uniforme staaf met lengte L en lyndigtheid λ . Bepaal die traagheidsmoment I om 'n as deur sy middelpunt $L/2$.

A thin uniform rod has a length L and line density λ . Determine the moment of inertia I about an axis through its center $L/2$.

$$\lambda = \frac{m}{L}$$

$$I = \int r^2 dm$$

$$\lambda = \frac{dm}{dL}$$

$$dm = \lambda dL$$

$$I = \int x^2 \lambda dL = \lambda \int x^2 dL = \lambda \left[\frac{x^3}{3} \right]_{-L/2}^{L/2}$$

$$I = \frac{\lambda}{3} \left[\left(\frac{L}{2} \right)^3 - \left(-\frac{L}{2} \right)^3 \right]$$

$$= \frac{\lambda}{3} \left[\frac{L^3}{8} - \left(-\frac{L^3}{8} \right) \right]$$

$$= \frac{1}{12} ML^2$$

(4)

- (g) Deur die teorie van parallele-asse te gebruik, toon aan dat die traagheidsmoment I om 'n as deur sy een punt gegee word deur:

By using the parallel-axis theorem, show that the moment of inertia I about an axis through its one end is given by:

$$I = \frac{1}{3} ML^2$$

(2)